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Topography of Arterial Circle of the Brain in Egyptian Spiny Mouse (*Acomys cahirinus*, Desmarest)

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With 3 figures

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Summary

Using stained acryl latex-injected techniques, the arterial circle of the brain in *Acomys cahirinus* Desmarest was studied. Results revealed an important individual variability of investigated structure. Three morphological variants were found: (1) the lack of typical arterial circle – opened in front and the back side, (2) partial opened at the back side, (3) completely closed arterial circle. This finding is opposed to many species of mammals, including rodents, and especially laboratory mouse. In our point of view, it seems to be a specific character.

Introduction

The blood supply to the brain has been interesting since the beginning of the 20th century (Hoffman, 1990), and it is a very important subject in many papers today. The authors are interested both in topography of the brain arteries and Willis's circle in different groups of taxonomy. A very important aspect for many investigators is a practical use of these results. Paemeleire et al. (2005) and Uchino et al. (2004) tried to explain inter-dependence between migraine and possible anomaly of arterial circles. Using 1D and 3D models, Moore et al. (2005) try to explain the clinical aspect of occlusion of afferent arteries and the lack of brain vessels. Most studies are connected with morphology and topography of the brain's base vessels. In this aspect, some species of big (including domestic) animals were investigated in the beginning, and later the advanced technique was used to study small mammals too. Nevertheless, there are few papers describing the basal arteries of the brain in rodents. Only Brown (1966) and Lee (1995) investigated these arteries in the rat and Wiland (1974) and Okuyama et al. (2004) in mouse and Firbas et al. (1973) in mouse, rat and hamster. In addition, there are some congress abstracts connected with Willis circle in small mammals (Kuchinka et al., 2004b).

Okuyama et al. (2004) studying arterial circle in mouse tried to explain the cases of human brain circulatory system. Our preliminary observations in Egyptian spiny mouse (Kuchinka et al., 2004a) showed the important individual variability in this species. Therefore, we decided to verify our previous studies. Moreover, our results can be used in comparative anatomy analyses in wider aspect of mammal's investigations.

Material and Methods

Investigations were performed on 12 adult individuals. The animals were conventionally reared and fed standard diet and tap water *ad libitum* (Brylińska and Kwiatkowska, 1996). The animals were killed with ether anaesthesia and injected by using stained acryl latex through the left heart ventricle. The heads of animal were fixed in 4% formalin and after the decalcification in 5% HNO_3 or 10% HCL, injected arteries of the base of the brain were exposed. For the terminology, Anonymous (1994) was used.

Results

The base arteries of brain in Egyptian spiny mouse showed an important individual variability. Three morphological variants were found: (1) the lack of typical arterial circle, opened in the front and the back side – five individuals, (2) partial opened at the back side, arterial circle – four individuals, (3) completely closed arterial circle – three individuals.

In the first, variant is the lack of typical arterial circle – opened in front and the back side was found. The arrangement of arteries is similar to asymmetric pyramid supplied with the carotid internal arteries at the back side (Fig. 1a,b).

The medial part of investigated structures is supplied with – middle cerebral arteries. This is one trunk or two connected arteries. They are divided on T-shape structure. The thicker trunk at the right side and thinner at the left (Fig. 1b). Moreover, two choroids arteries leaved circle at the left side. In the front part of the studied structure, three arterial vessels, two at the right and one at the left side were observed. They were situated in longitudinal cerebral fissure where they give anterior cerebral artery (nasal branches). The connection between the left and the right side was not observed.

At the back side, the basal artery of the brain supplies arterial circle (Fig. 1a,b). This artery gives the geminate nasal cerebellar arteries (thicker at the left side) and some thin midbrain arteries. Nasal cerebellar arteries are situated in transversal cerebral fissure. Just below, parallel with them, posterior cerebral arteries were found which are connected with communicating posterior arteries. Their connection with branches of basal artery was not observed.

In the second variant, the arterial circle is pyramid shape and opened at the back side (Fig. 2a). The main part of the circle is created with the connection of internal carotid artery, midbrain artery and anterior cerebral artery. At the right side meningeal branch was observed. In front part, at the left side, the corpus callosum artery leaved the arterial circle (Fig. 2b). In the beginning, this artery is parallel to anterior cerebral arteries where the small arterial connection was found.

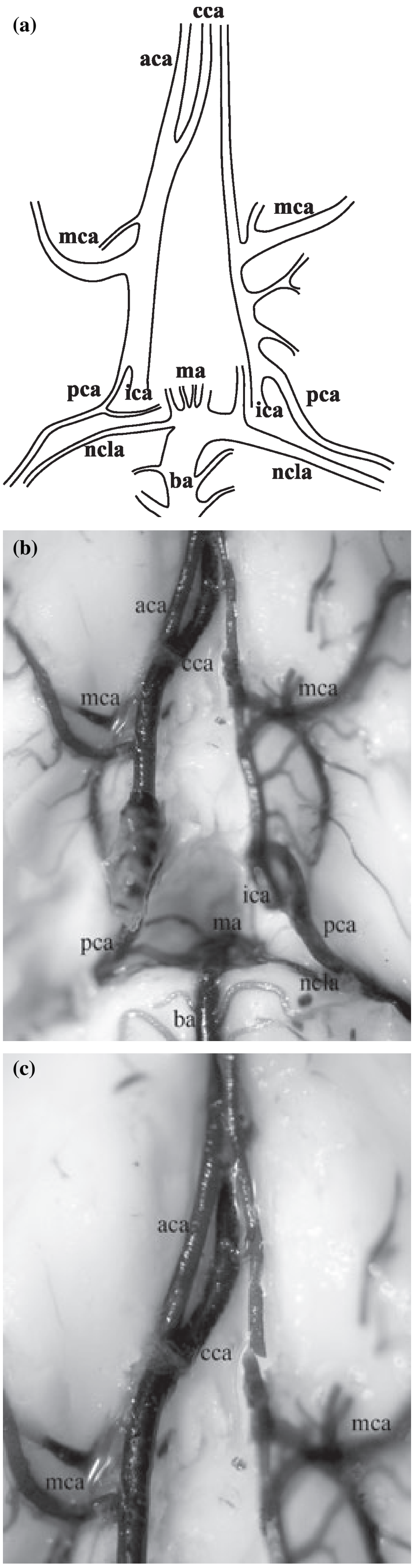

Fig. 1. Arterial circle of the brain in *Acomys cahirinus* Desmarest – variant 1. (a) Scheme of topography. (b) Arteries of Willi's circle. (c) Anterior part of arterial circle. aca, anterior cerebral artery (nasal branches); ba, basilar artery; cca, corpus callosum artery; ica, internal carotid artery; ma, midbrain arteries; mca, middle cerebral artery; ncla, nasal cerebellar artery; pca, posterior cerebral artery.

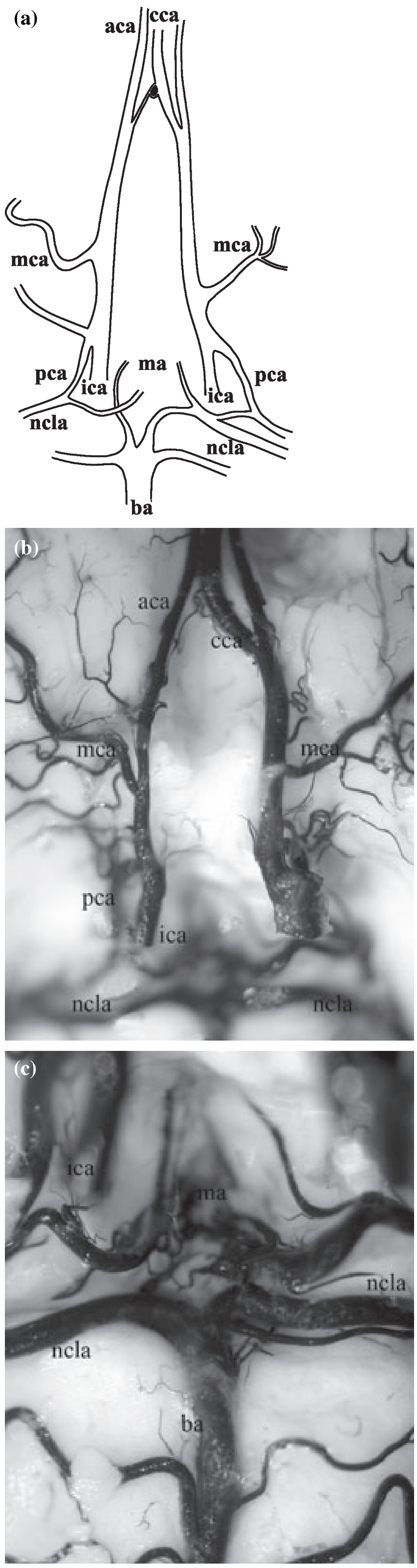

Fig. 2. Arterial circle of the brain in *Acomys cahirinus* Desmarest – variant 2. (a) Scheme of topography. (b) Arteries of Willi's circle. (c) Posterior part of arterial circle. aca, anterior cerebral artery (nasal branches); ba, basilar artery; cca, corpus callosum artery; ica, internal carotid artery; ma, midbrain arteries; mca, middle cerebral artery; ncla, nasal cerebellar artery; pca, posterior cerebral artery.

At the back side of arterial circle, the basilar artery gives at the right and the left side two symmetrical nasal cerebellar arteries and forward midbrain artery (Fig. 2a,c). The small connection between the left nasal cerebellar artery and the left posterior cerebral artery was found. At the right side, this connection was not observed.

In the third variant, the pyramid shape arterial circle is completely closed (Fig. 3a,b). Generally, the arrangement of arteries is similar to the previous described structure but both at the right and at the left side, the small connection between nasal cerebellar artery and posterior cerebral artery was observed. Moreover, in front part of arterial circle, the symmetrical anterior cerebral arteries were connected with uneven corpus callosum artery (Fig. 3c).

Discussion

As the results of our investigations showed, there is no uniform type of the brain arterial circle in Egyptian spiny mouse as opposed to many species of mammals (Brudnicki, 2000; Frąckowiak, 2003). In this species, the investigated structure was opened in the front and the back side, partially opened in the back side or completely closed.

The next of kind of presented species is domestic laboratory mouse (*Mus musculus*). As Okuyama et al. (2004) showed, in this species, only one type of arterial circle was observed. It was a structure opened at the back side, without the arterial connection between basilar artery and posterior cerebral artery. The rest pyramid shape arrangement of arteries is identical with this structure in Egyptian spiny mouse.

The review of literature indicates in many species of mammals, including rodents, typical kind of Willis circle. For example, typical arterial circle was found in guinea pig (Majewska-Michalska, 1997), chinchilla (Gielecki et al., 1996), guinea pig (Ocal and Ozer, 1992), cattle (Brudnicki, 2000), porcupine (Aydin et al., 2005), rhesus, goat and sheep (Kapoor et al., 2003).

More developed front part of arterial circle was observed in mouse and the main source of blood supplying of brain are internal carotid arteries (Firbas et al., 1973). Okuyama et al. (2004) describe a stronger development of front part of brain in mouse, particularly rhinencephalon, which must be properly supplied with blood. In Egyptian spiny mouse, arterial circle is supplied with internal carotid arteries and moreover in some morphological variants with basilar artery.

According to Frąckowiak (2003), in some family of rodents (Hystricidae, Cavidae, Hydrochoeridae, Dasyproctidae, Chinchilidae and Myocastoridae), extracranial part of carotid internal artery is reduced in obliteration process and the brain is supplied mainly with the basilar artery, assisted with maxillary artery, indirectly external ophthalmic artery connected with the arterial circle by the internal ophthalmic artery. The similar situation can be in ruminants where internal carotid artery is reduced and only in some species a very thin vessel was found (Godynicki, 1972).

According to literature data (Brown, 1966; Ocal and Ozer, 1992), in blood, supplying of arterial circle can be supremacy basilar artery, posterior cerebral artery or internal carotid artery. In two species examined rodents, in rat and guinea pig, Willis circle is formed from basilar artery and internal carotid artery. Gielecki et al. (1996) showed that diameter of basilar artery was bigger than all the rest of arteries in arterial circle in

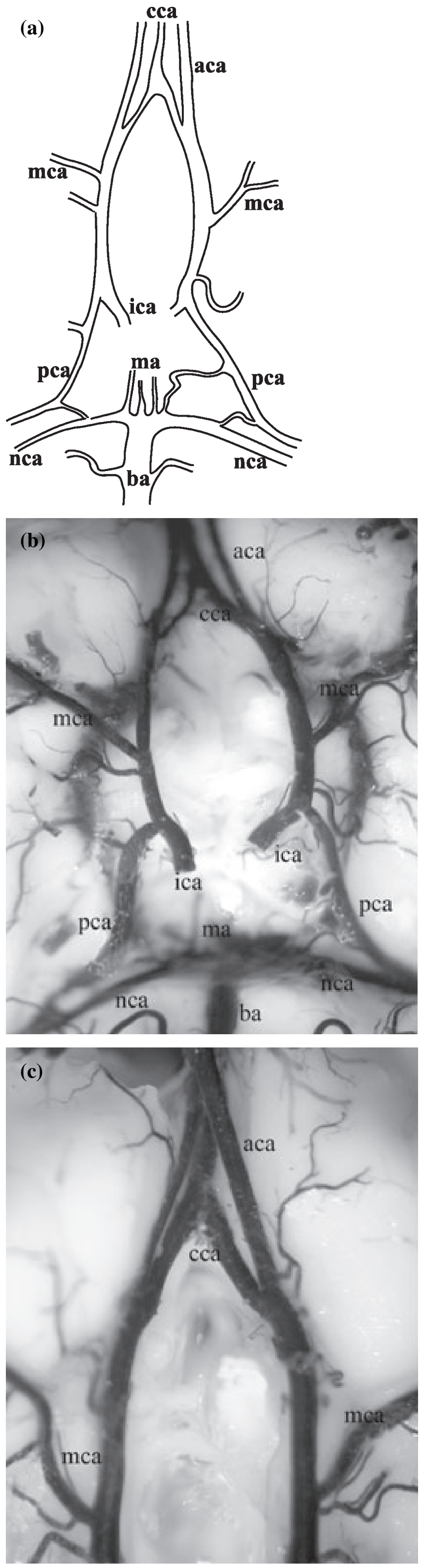

Fig. 3. Arterial circle of the brain in *Acomys cahirinus* Desmarest – variant 3. (a) Scheme of topography. (b) Arteries of Willis's circle. (c) Anterior part of arterial circle. aca, anterior cerebral artery (nasal branches); ba, basilar artery; cca, corpus callosum artery; ica, internal carotid artery; ma, midbrain arteries; mca, middle cerebral artery; ncla, nasal cerebellar artery; pca, posterior cerebral artery.

chinchilla. In this species, this artery gives the main source of blood supplying of the brain. Similarly, Goetzen and Sztumska (1992) showed that in human blood supplying of hippocampus is from posterior cerebral artery as well as in cat, rabbit and sheep. Moreover, Majewska-Michalska (1997) showed the main arteries of supplying thalamus are posterior cerebral artery and communicating artery posterior. In arterial circle of porcupine is supremacy of basilar cerebral artery (Aydin et al., 2005). According to Van Overbreeke et al. (1991) in human fetus, differently from adults, the occipital lobes are supplied by the internal carotid arteries and communicating arteries posterior. The similar conclusion was performed by Hillen et al. (1991) that compared diameter of Willis circle arteries and found a big variability of these vessels.

In our point of view, the wide variability in arrangement of Willis circle arteries in Egyptian spiny mouse seems to be a species character. Similar species characters were observed in mouse where midbrain artery is a single structure leaving internal carotid artery (Wiland, 1974) whereas in rat two separated vessels were observed (Brown, 1966; Firbas et al., 1973). It seems to be necessary to perform many comparative anatomy investigations in many species of rodents to confirm this thesis.

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